The Long Boil

Game Design Document · Final

A steampunk idle game · Jeremy Rule · 25 October 2026 · Capstone, Multi-Agent AI for Game Development

1 Executive Summary

The Long Boil is a steampunk idle MMO for web and phone with a mystery worth checking in for. Kettleport has ridden a column of steam a mile above a boiling sea for 312 years, and the Board of Gauges will not say why the sea is still boiling. You are a Tinker who inherits a cold workshop at No. 9 Sprocket Row from Hester Vane, who vanished with a notebook full of questions. Your Tinker keeps smelting, crafting and fighting for 12 hours while you are away. When you come back, the city has written to you.

At a glance	
Genre	Idle MMO with a storylet narrative: short scenes, chosen a few at a time
Platform	Web, as an installable app; iOS and Android later from the same Expo codebase
Session	3 to 5 minutes, 2 to 4 times a day
Capstone build	Prologue and Acts I and II across two zones (levels 1 to 45), 16 skills, the Brass Exchange, guilds and chat; Act III is the stretch
Playable now	The Prologue and most of Act I, on the dev build at thelongboil.com
Built by	Me, a Claude Code lead session, and six subagents

Core loop. Every check-in has two halves, and each feeds the other.

- **Idle:** collect what your Tinker made → craft it up a chain or sell it → gear up → queue the next 12 hours.
- **Story:** read the letters the Pneumatic Post delivered → spend Nerve on scenes whose options open as your skills rise → close the tab.

Levelling Tinkering is how you earn the right to pick Hester's lock, and 400 ingots smelted overnight bring a Stoker's letter asking for 20 of them.

- **Win:** an idle game never ends, so winning comes in layers. The capstone's finish line is concrete: defeat Old Bellows in the Cinderworks Foundry and translate its last breath, which closes Act II. Act I closes when you beat Scrapjaw, read the first page of Hester's notebook and choose one of four Ambitions. After that, players win personal progress (levels, gear, leaderboard ranks), a personal story (their Ambition's ending) and, each season, a city story (their guild's side carrying the Reckoning).
- **Lose:** a knockout (60-second respawn, then the queue resumes), a lost boss fight (the attempt is spent), a failed calibration (gear falls to +0 unless protected), or a redlined Needle: push one of four risk dials to 8 and the story sends you to the Infirmary, the Debtors' Gallery, the Valve Gaol or the Quiet Rooms until you talk, pay or think your way out. Story losses never stop the queue or touch your inventory.
- **Business:** free to play. Gilded Cogs, sold through Stripe (test mode for the capstone), buy time savers and cosmetics, never stats, Nerve or story.

Pillars. Progress while you're gone. Every skill opens a door. A city with a secret, told slowly. A real economy, where every roll can be replayed. Fair to players who never pay.

2 Game Mechanics

Everything here is something the player does in a click or two. The server does the arithmetic, online or off.

Action	What the player does and sees
Queue work	Pick an action and how long; the picker defaults to an hour, with quick buttons for more. The Boiler gauge shows the steam left, up to 12 hours. Two jobs fit in the queue: "smelt 400 ingots, then tinker 50 spanners".
Come back	A While You Were Away ledger lists what was made, fought and sold.

Craft and sell	Work up chains: dig copper, smelt Brass Ingots, tinker Brass Spanners. Sell to players on the Brass Exchange, an order book that fills even while both sides are offline, or to the Counting-House window at a fixed price.
Fight	Choose a zone, a loadout (one weapon, one armour piece) and a trigger. Watch hit sparks and damage numbers, or walk away. Each monster is weak to Melee, Marksmanship or Aetherics. A boss is its own queue item and allows 3 Rations, so Cooking matters from day one.
Read and choose	A scene is a card: a picture, at most 120 words and 2 to 4 options. Each option shows its Nerve cost and its odds before you commit.
Calibrate	Push gear from +1 to +20, odds shown first. Failure resets it to +0 unless a crafted Safety Valve limits the loss to one level.
Join in	City, Trade and Help chat, whispers, guilds of up to 30, and a Work Order every 8 hours carrying a faction's seal.

Your first twenty minutes

1. You sign in with Google or Discord. A solicitor's letter: No. 9 Sprocket Row is yours, contents included.
2. Professor Brassbuckle peels the Gauger tape off your Boiler. Its gauge fills to twelve hours, which is how long your Tinker works without you.
3. She leaves 50 Brass Ingots from the old stock. You queue 50 Brass Spanners and, like everyone, watch the first one land.
4. A canister rattles into the tray. It is Pip, with a bill for the workshop's back rent.
5. Rust Rats have been raiding the Row. You equip a spanner, set "eat a Ration below 40% HP", and win your first fight.
6. You sell the rat scrap at the Counting-House window and pay the bill.
7. Clearing the workbench drawer, you find a brass plate scratched with a gauge reading that does not match today's Gazette. Brassbuckle changes the subject.
8. You fill the queue and close the tab. Act I opens on your next visit.

Skills. Sixteen skills, each 1 to 100 on one shared XP curve: Prospecting, Lumbering and Scavenging gather; Smithing, Tinkering, Tailoring, Cooking and Tea Blending make; Calibrating improves; Vitality, Pressure, Precision, Melee, Fortitude, Marksmanship and Aetherics fight. Each is also a story stat: Smithing forces a door, Tinkering picks a lock, Tea Blending talks a deacon round.

Zone	Levels	Monsters	Boss and drop	Act
Rustmoor Scrapyards	1 to 20	Rust Rats	Scrapjaw: Scrapjaw's Cleaver, Ledger Page 1	I
Cinderworks Foundry	20 to 45	Soot Wraiths	Old Bellows: Bellows Plate, a Whistle notation	II
Thunderhead Reach	45 to 70	Storm Harpies	The Lightning Baron: Coil Staff	III (stretch)
Dreadnought Hangar	70+, 5 players	The Board's marines	The Iron Dreadnought: its armour set	After the capstone

The story half

- **Nerve** is the story budget: 10 at most, refilling 1 every 36 minutes, offline too. Most scenes cost 1. Act beats are gated to the zone's level band, so a two-week zone carries an act's worth of scenes.
- **The Pneumatic Post** delivers a letter every 2 hours and holds 3. A letter is a one-action scene: a bill, a summons, a neighbour who saw what your queue made.
- **Challenges** use one formula, printed on the button: `P = min(1, max(0.1, 0.6 × skill ÷ difficulty))`. Smithing 15 against difficulty 15 is 60%; Smithing 25 is certain.
- **The Needles** (Scald, Arrears, Notice and Haze) rise on risky choices and fall 1 a day. Haze stays hidden until Act II, when the Great Whistle starts sounding off the hour.
- **Hester's Ledger** collects her notebook pages from bosses, scenes and rare drops, with a pity timer that guarantees the next page within 1,000 kills in a zone.

Act I, as the player lives it (Rustmoor, levels 1 to 20)

1. The Gazette announces a routine review of the city's altitude. In the Lowvents, everyone knows "review" means a district gets dropped.
2. Hester's laboratory is sealed. A Tinkering check opens a drawer, and an index card inside reads "R.S. The Rats have it."
3. Scavenging the rats' nests turns up torn gauge charts thrown off Gaugehill.
4. You fix a vent for Mother Ash and take your own reading of the sea. It disagrees with the Gazette by 40 fathoms.
5. Inspector Rook starts turning up wherever you have just been. Cooperate, or watch Notice climb.
6. You beat Scrapjaw. Inside its chest is Ledger Page 1: the sea has risen 70 fathoms in 11 years, not the 6 the Gazette admits.
7. You give the page to the editor, the Inspector or the First Gauge, or keep it. Every road ends with a pass to the Foundry and your choice of Ambition.

A four-minute check-in, a week later

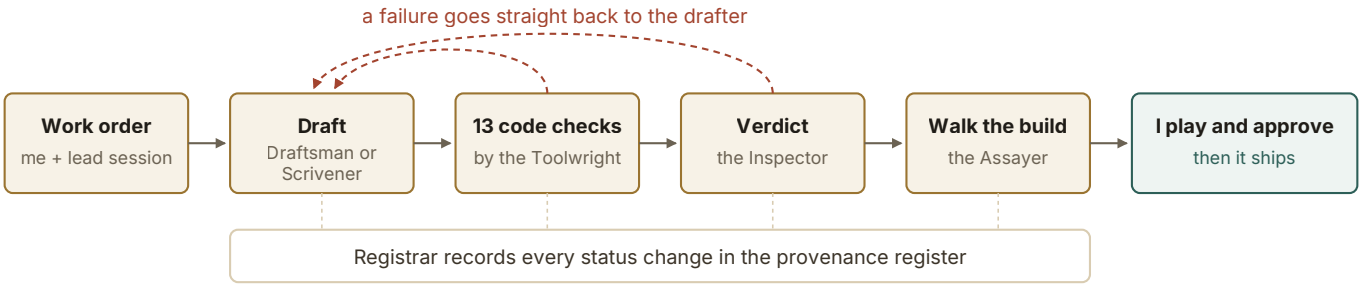
0:00	The ledger: 400 Brass Ingots smelted, 2 Rust Rat kills short of a level, one Exchange order filled.
0:30	Sell the spare ingots. Requeue Smithing, then Rustmoor.
1:00	Three letters. A Stoker wants 20 of those ingots, with a favour attached. Agree.
1:45	Spend 6 Nerve on Act I: two scenes in the Lowvents, one failed Smithing check (Scald rises to 2), one success.
4:00	Close the tab. Nothing needs you for six hours.

Economy. Calibrating is the main Crown sink, beside a 2% Exchange tax, the grocer's counter and guild founding (500,000 Crowns). NPC sell orders at a price ceiling keep a thin market moving. The Assayer owns the Crown budget, with a target that sinks remove at least 90% of what faucets add each week. Trading and Work Orders unlock at total level 50 on an account at least two days old, so throwaway accounts cannot farm them.

3 AI Architecture

Six Claude Code subagents build The Long Boil with me. Each is a Markdown file in the repository's *.claude/agents/* folder, owns files no other agent writes, and runs in its own context window. None runs while anyone plays: rolls, drops, fights and trades are deterministic server code, because idle play runs offline and a traded economy needs repeatable results.

Agent	Its job, in one sentence	What players notice
Draftsman	Drafts the game's numbers as JSON (monsters, drops, items, gear, recipes, tuning), deriving each value from the GDD or stopping to ask.	Rust Rats that take about a minute each at level 1, a Scrapjaw you cannot beat without Rations, and a first sale that pays the first bill.
Scrivener	Drafts scenes and Post letters in the house voice, using only the slice of canon it is cleared for.	Letters that react to last night's queue, and scenes that remember whether you kept the brass plate.
Toolwright	Writes the code checks every draft must pass before anyone reads it.	No scene that dead-ends, no reward that floods the market, no Act II secret turning up in Act I.
Inspector	Reviews what code cannot judge (voice, lore, names, reveal timing) and returns a verdict per item, never an edit.	Characters who stay in character, and a city that never explains its own joke.
Registrar	Keeps the provenance register: one row per content item showing who drafted it, what checked it and who approved it.	Mostly unseen: any bad item traces to its work order and comes out of the game quickly.
Assayer	Builds the game, walks it like a player, simulates pacing and Crowns, and reports what is playable and what needs deciding.	A check-in that always has something to do, and zones that take about two weeks.



One work order through the crew. Nothing reaches players without my sign-off.

Agents cannot leak what they were never told. The canon lives in git, sliced by act. A Scrivener cleared for Act I never receives the Act II truth, so it cannot hint at it even by accident. I write the story's spine myself (the Prologue, each act's climax, the Ambition openers and Hester's pages), and the agents write the ribs: district scenes, letters and repeatable scenes, which I edit.

> **The crew at work.** On 24 October the Assayer walked Act I through the live server and found it starved in the middle: between days 3 and 15, a player checking in three times a day got about 1.5 story actions a day and left 9.5 of 10 Nerve unspent. I set a target of 6 to 8. Work order 023 asks the Scrivener for 8 repeatable scenes, the Draftsman checks their rewards are not a Crown faucet, and the Assayer re-walks the act to prove the number. For the player, it is the difference between a check-in with nothing to read and one with a choice waiting.
>
> **Pipeline in one sentence.** In my game, the Scrivener takes my work orders and the Act I canon slice and produces scene JSON that the player sees as new letters in the Pneumatic Post and the next step on Hester's trail.

4 Technical Strategy

Built so far	As of 25 October 2026
Engine	The Boiler and offline resolution, combat and drops, the Exchange and Counting-House window, the story engine (Nerve, scenes, the Post, Needles, Hester's Ledger), chat, guilds and Work Orders: 11 milestones, 15 test files
Rustmoor content	30 actions, 41 items, 13 gear pieces, 5 monsters including Scrapjaw
Story content	27 scenes, 8 letters and a first-session guide for the Prologue and Act I; 16 Work Orders
Checks and register	13 code checks, all passing; 136 items in the register, 69 approved by me after playing them
Hosting	Live at thelongboil.com with Google and Discord sign-in, deployed from GitHub on every push

Stack. One TypeScript codebase. The client is Expo (React Native Web) as an installable web app. The server is authoritative Node.js on ECS Fargate. Every change to a Tinker is a row in one ordered event log in Aurora PostgreSQL, so seeded rolls plus a versioned content set make any result replayable, offline hours included. Sign-in is Cognito, behind CloudFront, and the environment is defined in AWS CDK.

Token budget. A player action costs 0 tokens, because no agent runs per player. The cost is building the game. These are estimates for each remaining week, scaled from the first week's 23 work orders and 11 engine milestones.

Agent	Owns	Model	Tokens / task	Tasks / week	Tokens / week
Lead session	engine code, work orders	Opus 5.5	n/a	5 days	10,000,000
Draftsman	items, monsters, drops, tuning	Sonnet 5.5	120,000	3	360,000
Scrivener	scenes, letters	Opus 5.5	100,000	3	300,000
Toolwright	checks, schemas, CI	Sonnet 5.5	150,000	2	300,000
Inspector	verdicts only	Haiku 5.5	40,000	6	240,000
Registrar	the provenance register	Haiku 5.5	20,000	8	160,000
Assayer	simulations, play reports	Sonnet 5.5	120,000	3	360,000
Total per week					about 11.7M

A Scrivener task is about 40K tokens a call (a 27K cached prefix of style guide, canon slice and quality list, plus the work order and neighbouring scenes) over two or three calls with revisions. About 80% of all tokens are cached reads. At October 2026 list prices that is about $21 a week, or about $105 over the five weeks to the capstone, under a $150 cap; on a Claude plan the same work counts against the plan's usage limits instead. After the capstone, the daily Brass Gazette adds about 14K tokens a day on Haiku 5.5.

Constraint	Impact and response
Output reliability (the biggest)	One bad number in a traded economy crashes prices players spent real time on, and one character who knows too much spoils the mystery for everyone at once. Code checks run first, the Inspector second, and I play every build before it ships.
Shared usage limits	Every subagent's requests count against the same limits as the lead session. Dependent work runs one agent at a time, prompts stay short, and bookkeeping goes to Haiku.
No shared memory	A subagent sees none of my conversation, so the GDD, canon slices, work orders and register live in the repository. That is also what makes need-to-know canon work.
Model inheritance	An agent file with no model line runs on the lead session's model, which is true of all six today. Pinning each to the model above is the cheapest saving available.
Haiku 5.5 price step	Haiku charges 5 times more above 100K tokens, so Inspector prompts stay under 60K.

When	Milestone	Playable result
Week of 26 Oct	8 repeatable scenes; my Prologue, Act I's close and the Ambition openers; Calibrating merged	Act I end to end, from sign-in to Ambition
Week of 2 Nov	Cinderworks Foundry: Soot Wraiths, Old Bellows, Tailoring and armour; leaderboards	Fighting past level 20
Week of 9 Nov	Act II story: Haze, Whistle notations, Old Bellows' last breath	The capstone finish line
Week of 16 Nov	The Assayer re-walks both acts; classmates playtest; a restore-from-backup drill	A stable demo build
24 Nov	Capstone demo	

If time runs short, cuts go in this order: Act III (already a stretch), the test-mode Cog Shop, guild charters and Guild Contracts, then Act II's district scenes. The idle loop, the Prologue and Act I are never cut; they are already playable. After the capstone come Act III, the five-player Dreadnought Hangar, seasons and the Reckoning, live payments and phone apps.

Version	Date	What it added
First draft	16 Oct 2026	The idle loop, five planned agents, a patch token budget
Second draft	25 Oct 2026	The story, win and loss conditions, the six-agent crew, a build token budget, dated milestones
Final	25 Oct 2026	The first session and Act I as the player lives them, the crew's pipeline, the state of the build

Sources: Claude Code docs, "Create custom subagents" (code.claude.com/docs/en/sub-agents) · Anthropic pricing, checked October 2026 · The Long Boil repository: docs/gdd.md, docs/build-plan.md, docs/work-orders/, content/provenance.json